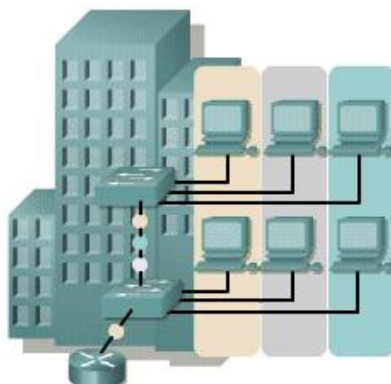


Virtual LANs

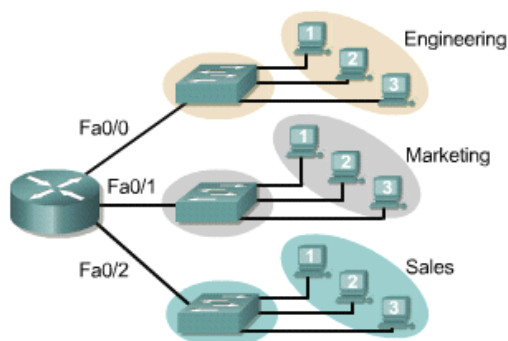
Miguel Leitão, 2018

VLANs

- Estruturam a rede de forma independente da estrutura física.
- Permitem agrupar utilizadores em função de políticas específicas, independentes da localização física.



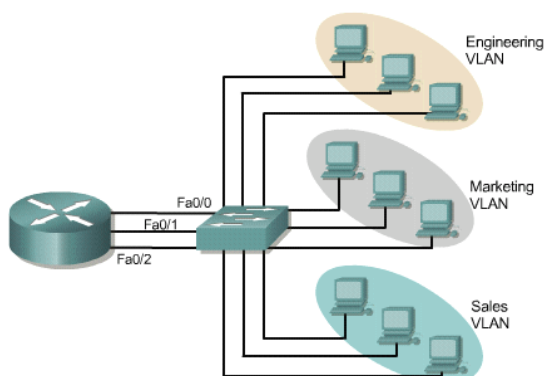
Segmentação tradicional



Sem VLANs:

Domínios de difusão são definidos pela estrutura física da rede, normalmente em função da localização geográfica.

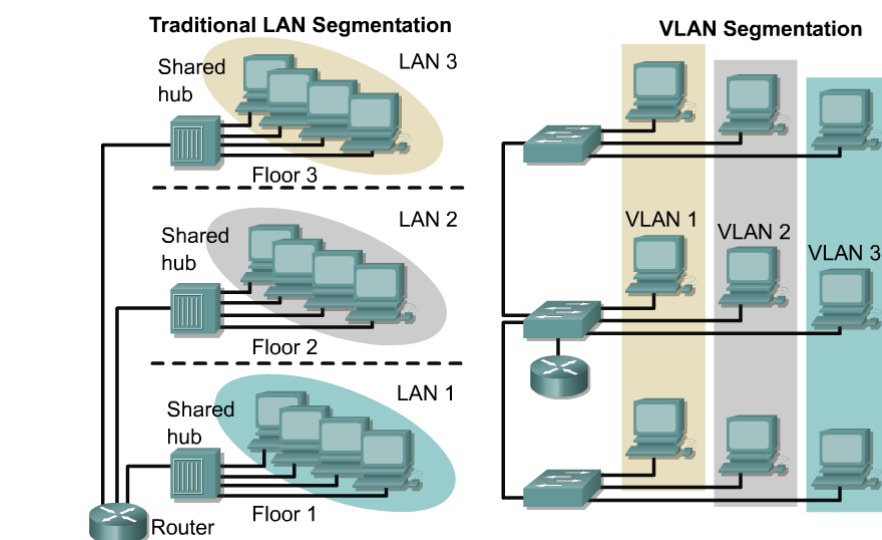
VLANs



Com VLANs:

Domínios de difusão podem ser definidos administrativamente, sem restrições da estrutura física ou da localização geográfica.

VLAN Segmentation



Benefícios das VLANs

- Facilita a instalação de postos de trabalho.
- Facilita a mobilidade de postos de trabalho.
- Facilita a estruturação lógica da rede.
- Melhora o controlo de tráfego.
- Aumenta a segurança.

Tipos de VLANs

- Port-based (static)
 - Tipologia mais usada.
 - Configuração simples.
 - Portas são atribuídas individualmente ou em grupos.
- MAC address
 - Raramente utilizada.
 - Administração pesada.
 - Cada endereço MAC deve ser configurado individualmente.
 - Permite soluções dinâmicas.
- Protocol based
 - Pouco comum.
 - Associa endereços IP ou gamas de endereços.

Static VLANs

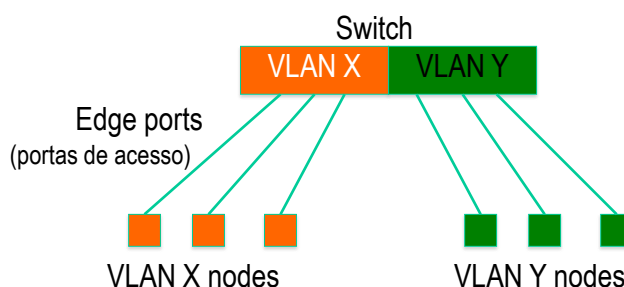
- Static VLANs work well when:
 - Moves are controlled and managed.
 - There is robust VLAN management software to configure the ports.
 - It is not desirable to assume the additional overhead required when maintaining end-station MAC addresses.

Local VLANs

- 2 VLANs or more within a single switch.
- Edge ports, where end nodes are connected, are configured as members of a VLAN.
- The switch behaves as several virtual switches, sending traffic only within VLAN members.
- Switches may not bridge any traffic between VLANs, as this would violate the integrity of the VLAN broadcast domain.
- Traffic between VLANs must be routed.

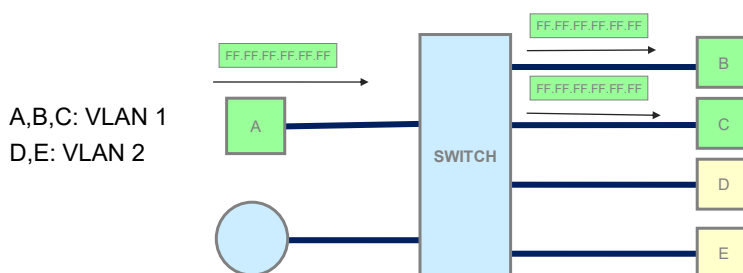
Local VLANs

- Allow splitting switches into separate (virtual) switches.
- Only members of a VLAN can see that VLAN's traffic
 - **Inter-VLAN traffic must go through a router.**



LANS Virtuais

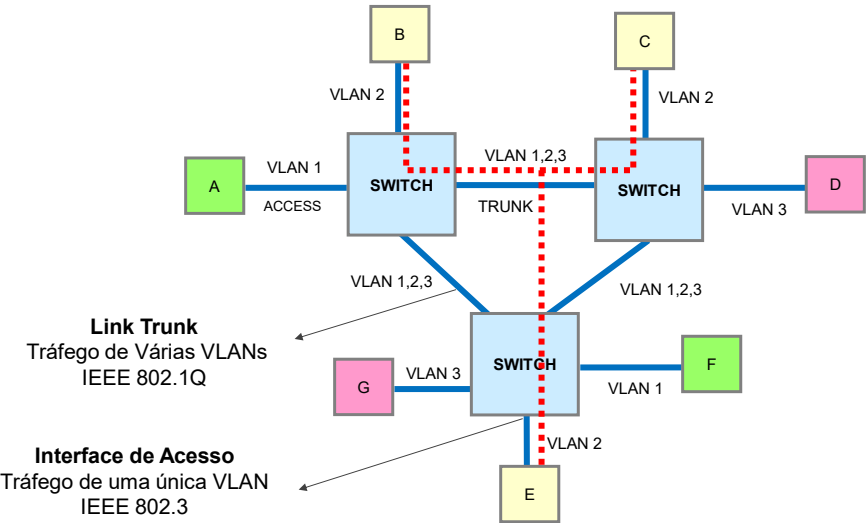
- SEGMENTO = Domínio de Colisão
 - Os computadores de um Hub estão no mesmo segmento.
- VLAN = Domínio de Broadcast
 - O tráfego de broadcast pode passar de uma VLAN para outra apenas através de um encaminhador.



VLAN operation

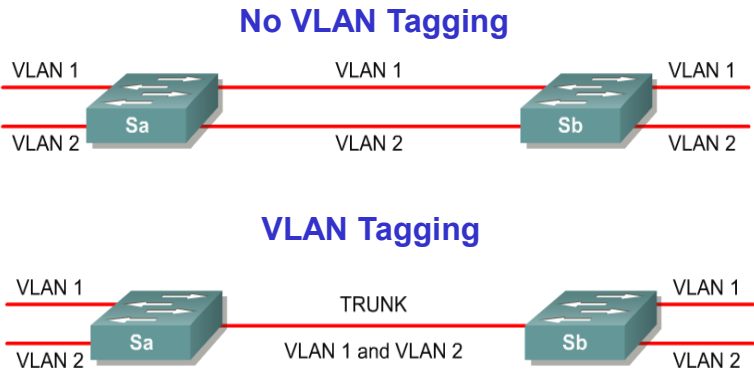
- As a device enters the network, it automatically assumes the VLAN membership of the port to which it is attached.
- The default VLAN for every port in the switch is VLAN 1 and cannot be deleted.
- All other ports on the switch may be reassigned to alternate VLANs.

Interligação de Switches



VLANs across switches

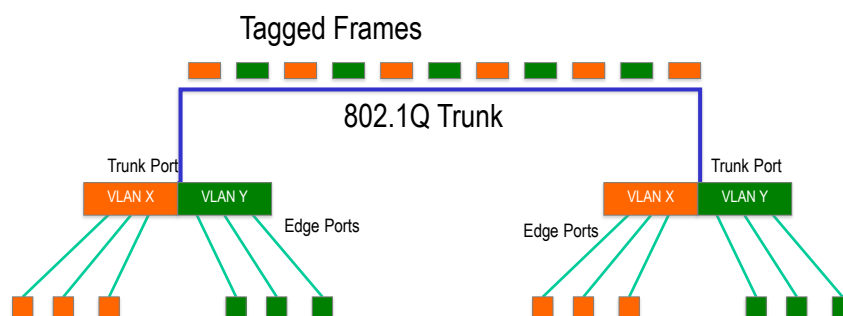
- VLAN tagging is used when a single link needs to carry traffic for more than one VLAN.



VLANs across switches

- Two switches can exchange traffic from one or more VLANs.
- Inter-switch links can be configured as **trunks**, carrying frames from all or a subset of a switch's VLANs.
- Each frame on a **trunk** carries a **tag** that identifies which VLAN it belongs to.

VLANs across switches



This is called **"VLAN Trunking"** or **"VLAN Tagging"**

Inter-Switch Link

Tagging	Method	Media	Description
Inter-Switch Link (ISL)	Fast Ethernet	ISL header encapsulates the LAN frame and there is a VLAN ID field in the ISL header	Frame is lengthened
802.IQ	Fast Ethernet	IEEE defined Ethernet VLAN protocol	Header is modified
802.IQ	FDDI	IEEE defined standard: The 802.10 protocol incorporates a mechanism whereby LAN traffic can carry a VLAN identifier	VLAN ID is the essential piece of required header information.
LAN Emulation (LANE)	ATM	No tagging	Virtual connection implies a VLAN id.

Quadros Ethernet

Ethernet I & II

>= 1536

MAC origem (6 bytes)	MAC destino (6 bytes)	Tipo Proto. (2 bytes)	Dados (46 a 1500 bytes)	CRC (4 bytes)
-------------------------	--------------------------	--------------------------	----------------------------	------------------

IEEE 802.3

< 1536

MAC origem (6 bytes)	MAC destino (6 bytes)	Tamanho (2 bytes)	Dados (46 a 1500 bytes)	CRC (4 bytes)
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IEEE 802.1Q

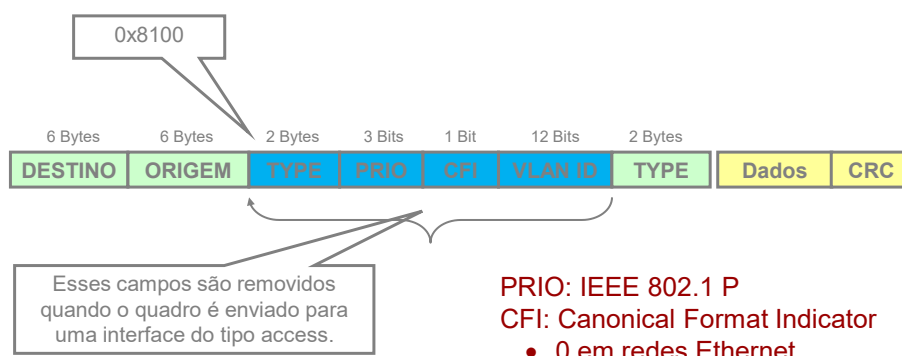
MAC origem (6 bytes)	MAC destino (6 bytes)	Tipo Proto (2 bytes)	VLAN Tag (2 bytes)	Tipo Proto (2 bytes)	Dados (46 a 1500 bytes)	CRC (4 bytes)
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Tipo 802.1Q = 0x8100

Prioridade (3 bits) + CFI (1bit) + VLANID (12 bits)

Protocolos Trunk

- Os quadros nas interfaces Trunk utilizam o formato IEEE 802.1Q.
- Cada quadro inclui identificador da VLAN a que pertence.



IEEE 802.1Q

- IEEE standard.
- Defines how Ethernet frames should be tagged when moving across switch trunks.
- Switches from different vendors are able to exchange VLAN traffic.

Modos das Portas de Switch

- As portas de um switch pode trabalhar nos modos:
 - **Modo Access**
 - Cada porta do switch pertence a uma única VLAN.
 - Quadros Ethernet: Formato Normal.
 - **Modo Trunk**
 - Multiplexa o tráfego de múltiplas VLANs.
 - Normalmente interconectam switches.
 - Quadros Ethernet: formato especial (VLAN).
 - Apenas computadores com placas especiais podem ser conectados.

Tagged vs. Untagged

- Edge ports are not tagged,
they are just “members” of a VLAN
- Frames in switch-to-switch links (trunks), when
transporting multiple VLANs, need to be tag.
- A trunk can transport both tagged and untagged
VLANs
 - **As long as the two switches agree on how to handle
those**

Link Aggregation

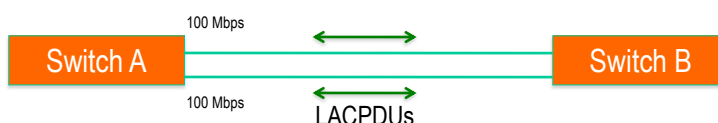
Link Aggregation

- Also known as **port bundling**, **link bundling**
- Multiple links can be used in parallel as a single, logical link
 - **For increased capacity**
 - **For redundancy (fault tolerance)**
- **LACP** (Link Aggregation Control Protocol) is a standardized method of negotiating these bundled links between switches

LACP Operation

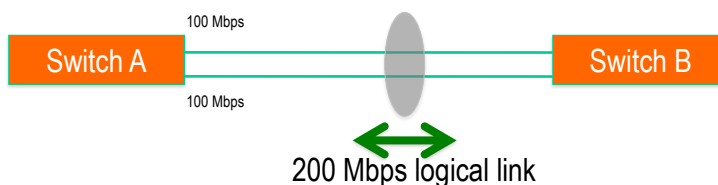
- Two switches connected via multiple links will send LACPDU packets, identifying themselves and the port capabilities.
- They will then automatically build the logical aggregated links, and then pass traffic.
- Switch ports can be configured as active or passive.

LACP Operation



- Switches A and B are connected to each other using two sets of Fast Ethernet ports
- LACP is enabled and the ports are turned on.
- Switches start sending LACPDU, then negotiate how to set up the aggregation.

LACP Operation



- The result is an aggregated 200 Mbps logical link
- The link is also fault tolerant: If one of the member links fail, LACP will automatically take that link off the bundle, and keep sending traffic over the remaining link.

Distributing Traffic in Bundled Links

- Bundled links distribute frames using a hashing algorithm, based on:
 - **Source and/or Destination MAC address**
 - **Source and/or Destination IP address**
 - **Source and/or Destination Port numbers**
- This can lead to unbalanced use of the links, depending on the nature of the traffic.
- Always choose the load-balancing method that provides the most distribution.